

For full credit explain your reasoning, showing all relevant work.

Exercise 1. Consider a permutation $\alpha \in S_5$ which is a product of two cycles:

$$\alpha = (1, 2, 3, 5) \cdot (4, 1, 3)$$

- a) Write α in the matrix notation.
- b) Write α as a product of disjoint cycles.
- c) Compute the order of α .
- d) Write α^{20} in the matrix notation.

Exercise 2. Compute all possible orders of elements in S_7 .

Exercise 3. Compute the number of elements of order 12 in S_7 .

PRACTICE PROBLEMS

Exercises below are for practice only - do not turn them in for grading.

Practice Exercise 1. Let $\beta \in S_6$ be a permutation given as the following product of cycles

$$\beta = (1, 4, 5, 6, 2) \cdot (2, 3, 4, 5) \cdot (1, 3, 6) \cdot (2, 3, 5)$$

Write β^{10} as a product of disjoint cycles.

Practice Exercise 2. Compute the number of elements of order 6 in S_7 .

Practice Exercise 3. Give an example of a permutation which is an element of order 30 in the group S_{12} .

Practice Exercise 4. For $\alpha \in S_n$, show that $|\alpha|$ divides $n!$.